Radiation

Exam code: 0625 & 0972

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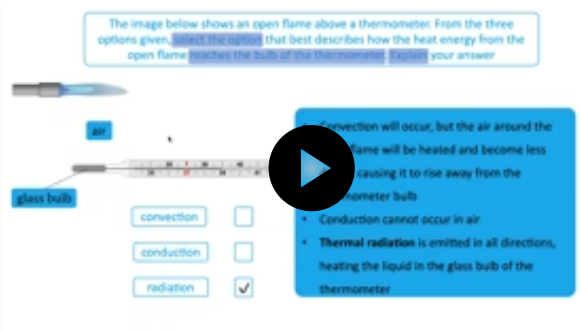

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Thermal radiation

What is thermal radiation?

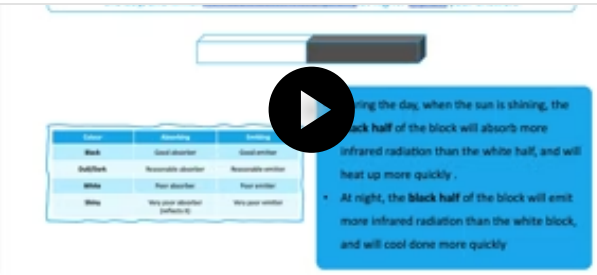
- All objects give off thermal radiation
  - The hotter an object is, the more thermal radiation it emits
  - Thermal radiation is **infrared radiation** (part of the electromagnetic spectrum)
- Thermal radiation is **the only way in which heat can travel through a vacuum**
  - It is the way in which heat reaches us from the Sun through the vacuum of space
  - Thermal radiation does **not need a medium** to travel, unlike convection and conduction

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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Effects of different surfaces

What two factors affect the rate of thermal energy transfer?

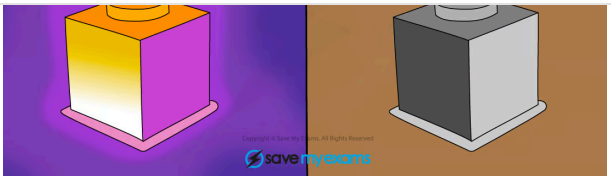
- The amount of thermal radiation emitted by an object depends on a number of factors:
  - The **surface colour** of the object (black = more radiation emitted and absorbed)
  - The **texture** of the surface (shiny surfaces = less radiation emitted and absorbed)

Table of the effect of coloured surfaces on absorbing and emitting ability

Colour/texture	Absorbing	Emitting
Black	Good absorber	Good emitter
Dull/dark	Reasonable absorber	Reasonable emitter
White	Poor absorber	Poor emitter
Shiny	Very poor absorber (it reflects)	Very poor emitter

- Black objects are very good at absorbing thermal radiation, for example black clothes make you feel hotter in sunny weather
  - Black objects are also very good at emitting thermal radiation, which is the reason that some chargers for laptops, and radiators in cars are coloured black - it helps them to cool down

- Shiny objects reflect thermal radiation and so absorb very little
  - They also emit very little, though, and so take longer to cool down

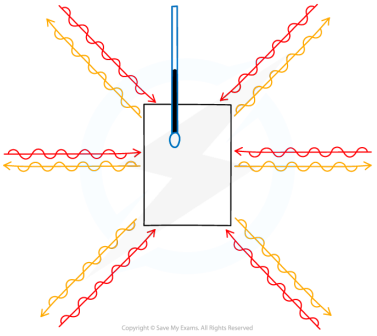


The infrared radiation emitted from a hot object can be detected using a special camera. The dull black side of the cube (left) is seen to glow brighter with infrared radiation than the shiny light side of the cube (right)

Thermal equilibrium

Extended tier only

- As an object **absorbs** thermal radiation it will become **hotter**
  - As it gets hotter it will also **emit** more thermal radiation
- Eventually, an object will reach a point where it is **absorbing** radiation at the **same rate** as it is **emitting** radiation
  - At this point, the object will be in **thermal equilibrium**
  - At thermal equilibrium, an object has **constant temperature**



An object will remain at a constant temperature if it absorbs heat at the same rate as it loses heat

Heating & cooling by energy transfer

Extended tier only

- If the rate at which an object **receives** energy is **greater** than the rate at which it **transfers** energy **away**:
  - then the object's temperature will **increase**
- If the rate at which an object **transfers** energy **away** is **greater** than the rate at which it **receives** energy:
  - then the object's temperature will **decrease**

- As temperature decreases, the object transfers less energy until the rates are equal

Examiner Tips and Tricks

When a question on heat transfer mentions the surface properties of an object, such as describing it as shiny, black or white, then you are being clued-in to write about **thermal radiation**.

Shiny things are both poor emitters **and** poor absorbers of thermal radiation, while black surfaces are both good emitters **and** good absorbers.

The surface makes the object either good or bad at **both**, so you don't need to remember too many facts!

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